

Non-Verbal Analogies

COMPLETE LEARNING GUIDE

What You'll Learn

Non-verbal analogy questions (Figure A is to Figure B as Figure C is to ?) require identifying the exact transformation that turns A into B and then applying that identical transformation to C to derive the answer. This sub-module goes deeper than the brief analogy coverage in 'Pattern recognition' by working through every major analogy sub-type — single transformations (rotation, reflection, resize), element addition/removal counted precisely, shading and fill-position changes, two-step combined transformations, and categorical (non-transformational) analogies based on a shared property rather than a geometric change — plus a verification step for when a derived answer doesn't match any given option.

1. The Basic Method: Derive the Rule From A:B First

Every non-verbal analogy is solved by first determining the EXACT transformation that turns figure A into figure B, stated explicitly (not just 'it changed somehow'), and only then applying that same exact transformation to figure C to derive the answer — never look at the answer options before pinning down the A:B rule precisely.

- Figure A is a small unshaded triangle; figure B is a large shaded triangle (same shape, but bigger and now filled in). The A:B rule: 'increase size AND add full shading.' Figure C is a small unshaded square. Applying the identical rule: the answer must be a large, fully shaded square.

2. Single-Transformation Analogies: Rotation and Reflection

The simplest analogies change exactly one geometric property between A and B — most often a rotation by a fixed angle or a left-right/top-bottom flip — identify the transformation type and its exact amount (the angle, or which axis it's flipped across) before applying it to C.

- Figure A is an L-shape with its corner at the bottom-left. Figure B is the same L-shape rotated 90° clockwise (corner now at top-left). The rule: 'rotate 90° clockwise.' Figure C is a T-shape with its stem pointing down. Applying the same rotation: the answer is the T-shape rotated 90° clockwise, with its stem now pointing left.

3. Element Addition or Removal Analogies

Some analogies keep the outer shape constant but change the number of internal elements (dots, small lines, smaller embedded shapes) between A and B — count the exact change (added 2 dots, removed 1 line) rather than just noting 'more' or 'fewer,' since the exact count must transfer precisely to C.

- Figure A is a circle with 1 dot inside; figure B is the same circle with 3 dots inside (an increase of exactly 2 dots). Figure C is a square with 2 dots inside. Applying 'add exactly 2 dots': the answer is a square with 4 dots inside — not just 'more dots than C,' but precisely 2 more.

4. Shading and Fill-Pattern Analogies

Shading-based analogies change which portion of a figure is filled between A and B — describe the shading change in exact positional terms (which specific section becomes shaded) rather than just 'shading changed,' since the position of the new shading must also transfer correctly to C.

- Figure A is a circle with its left half unshaded; figure B is the same circle with its RIGHT half now shaded instead (the shading moved from left to right, a 180° shift). Figure C is a square with its top half shaded. Applying 'shade shifts 180° (opposite side)': the answer is a square with its bottom half shaded instead.

5. Two-Step Combined Analogies

Higher-difficulty analogies apply two independent transformations from A to B simultaneously (such as a rotation AND an element count change together) — isolate and state each transformation separately, then apply both, independently, to figure C rather than trying to see the combined change as a single vague impression.

- Figure A is an arrow pointing up with 1 star beside it. Figure B is the same arrow rotated 90° clockwise (now pointing right) with 2 stars beside it. Two rules: 'rotate 90° clockwise' AND 'add 1 star.' Figure C is an arrow pointing left with 3 stars. Applying both rules independently: rotate 90° clockwise (arrow now points up) AND add 1 star (4 stars total) — the answer is an up-pointing arrow with 4 stars.

6. Categorical (Non-Transformational) Analogies

Not every analogy is a geometric transformation — some pair figures by a shared CATEGORY instead, such as 'both are made only of curved lines,' and the answer must belong to the same category as C rather than being a transformed version of C's specific shape.

- Figure A (a circle) relates to figure B (an oval) by the shared category 'made only of curved lines, no straight sides.' Figure C is a wavy S-shape (also purely curved). Since no rotation or resize rule fits A!B at all, the correct approach recognizes this as a category match: the answer must be any shape made only of curved lines, not a specific transformed version of C.

7. Verifying the Rule Against the Options, Not Just Deriving It

After deriving a rule from A:B and applying it to C, always check the derived answer against the actual answer options provided — if no option matches exactly, the originally derived rule was likely wrong or incomplete, and the A:B pair should be re-examined for an additional property that was initially overlooked.

- Derived rule from A:B: 'rotate 90°.' Applying to C predicts a specific rotated figure, but no answer option matches it exactly — re-examining A and B reveals a SECOND change also occurred (the shape also got slightly larger), a detail missed on the first pass. Reapplying both 'rotate 90°' AND 'enlarge' to C now matches one of the given options correctly.

8. Worked Full Example: A:B::C:? With Two Properties

Combine the full method: state the rule from A to B precisely (including every property that changed, not just the most obvious one), apply every part of that rule to C, and confirm the result against the answer choices before finalizing.

- A: a small unshaded pentagon. B: a small SHADED pentagon rotated 72°. Two changes: 'add full shading' AND 'rotate 72°.' C: a large unshaded hexagon. Applying both changes: shading gives a large shaded hexagon; the 72° rotation is then applied to that shaded hexagon too. The final answer is a large, fully shaded hexagon rotated 72° from C's original orientation.

Common Pitfalls

- Looking at the answer options before precisely deriving the exact A!B transformation rule.
- Stating a rule vaguely ('it got different') instead of naming the exact rotation angle, element count change, or shading position.
- Missing that an analogy combines TWO transformations at once and applying only the more obvious one.
- Assuming a categorical (property-based) analogy must be a geometric transformation, and searching for a nonexistent rotation/resize rule instead.
- Not re-checking a derived answer against the actual options, and missing a second overlooked transformation as a result.
- In element-count analogies, noting only the direction of change ('more dots') instead of the exact number changed.
- In shading analogies, describing the change only as 'more shaded' instead of specifying exactly which section became shaded.

Key Takeaways

- Always derive the exact A!B transformation rule first, stated precisely, before looking at any answer option.
- Check for combined (two-property) transformations — rotation AND element count, or shading AND size — rather than assuming just one change.
- Categorical analogies group by a shared property, not a transformation; don't force a rotation/resize rule where none exists.
- State element-count and shading changes with exact numbers and positions, not vague direction words.

- If a derived answer doesn't match any option, re-examine A:B for a second, overlooked property before concluding the question is flawed.
- Apply every part of a multi-part rule independently to C, then combine the results into one final figure.

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GeeksforGeeks — Non-Verbal Reasoning Questions and Answers

<https://www.geeksforgeeks.org/non-verbal-reasoning-questions-and-answers/>